EE-482: Network Security and Cryptography

LECTURES NO 4:

i. *Cryptanalysis*
ii. *Symmetric Key Cryptography*
iii. *Asymmetric Key Operation*

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Organization of Lecture

1. Cryptanalysis
2. Types of Attacks
3. Some concepts on Encryption
4. Comparison between Link-to-Link and End-to-End Encryption
5. Examples on calculating decryption time(s) for an exhaustive search using different key sizes
6. Symmetric Key Cryptography
7. Examples on Diffie-Hellman
8. Asymmetric Key Operation

1. Cryptanalysis (1/4)

* **Cryptanalysis:**
  - → cost to break: value of information
  - → time to break: lifetime of information

Discover key and Plaintext

Assume: algorithm is known

Security of a cryptosystem doesn't depend on keeping secret the algorithm ⟹ Security depends on keeping secret the key: Kerckhoff's Principle.

1. Cryptanalysis (2/4)

* **Key size:**

↳ A brute force approach involves trying every possible key until intelligible translation of cipher text into plain text is obtained

↳ Key size refers to number of bits that make up the key. eg: 32 bits, 56 bits, 128 bits etc.

↳ Larger the key size the more difficult it is to crack the key using brute force attack.

→ 1 bit key   0 or 1

→ 2 bit key   00 or 01 or 10 or 11

Average time required for exhaustive key search (Brute Force Attack)

- Always possible to simply try every key
- Most basic attack, proportional to key size

Key Size (bits)	Number of Alternative Keys	Time required at 10^6 Decryption/µs
32	$2^{32} = 4.3 \times 10^9$	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	10 hours
128	$2^{128} = 3.4 \times 10^{38}$	5.4×10^{18} years
168	$2^{168} = 3.7 \times 10^{50}$	5.9×10^{30} years

2. Types of Attacks (1/2)

*** Types of Attacks**

1). Cipher text only attack:

↳ attacker has no clue about the plain text.
↳ can use brute force (try all possible keys)
↳ if key size is larger this becomes impractical.
↳ attacker can rely on analysis of cipher text (make use of knowledge of language e.g: frequency of various letters)

2) Known plain text attack:

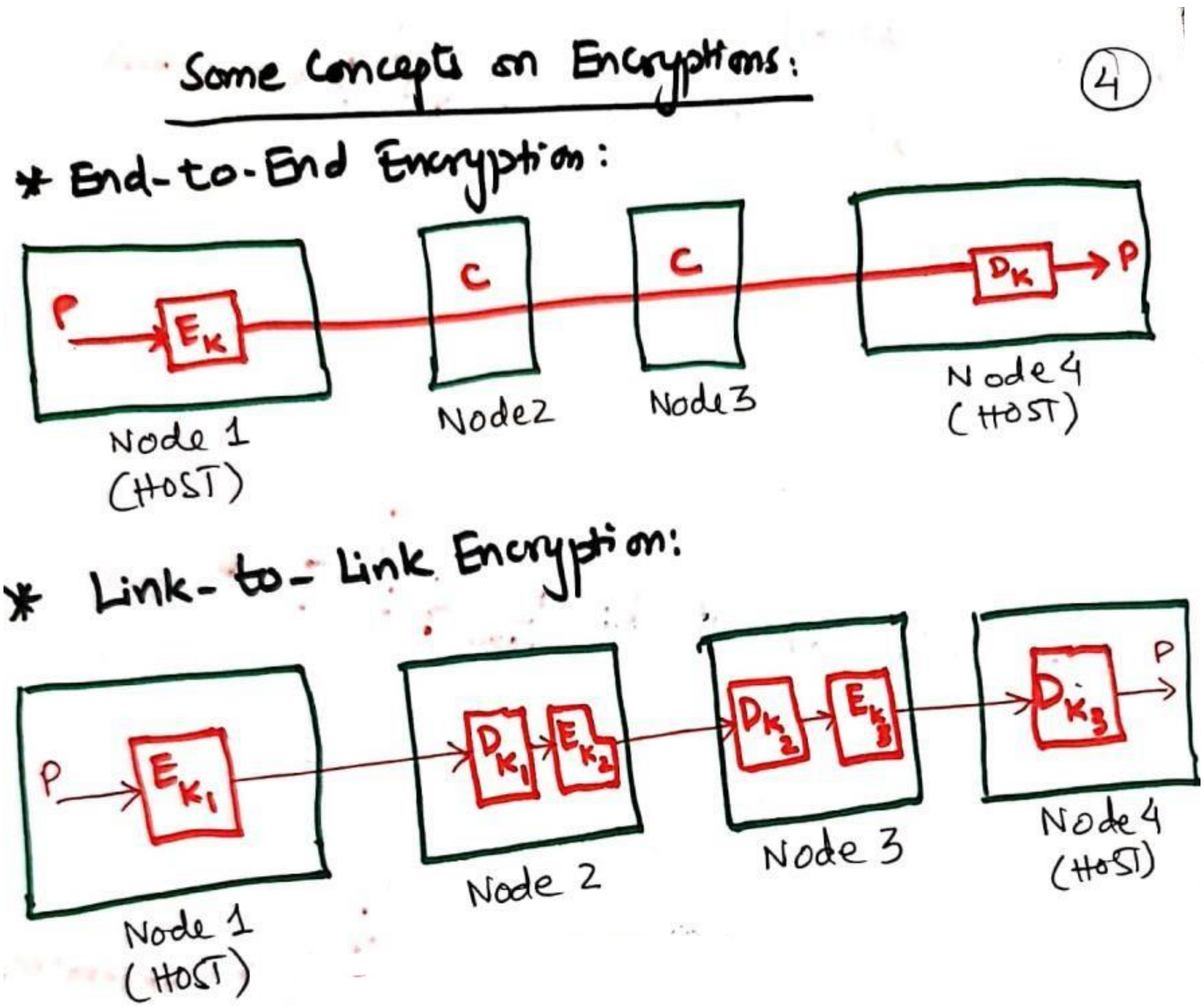
↳ attacker knows about some pairs of plain text & corresponding cipher text.
↳ attacker tries to find out more pairs and thus gets to know more & more plain text.

2. Types of Attacks (2/2)

3) Choosen plain text attack: $P_{known} \rightarrow \boxed{E} \rightarrow C_{generated}$.

- ↳ Attacker is able to some how get to the source system to insert into the system a message choosen by the attacker.
- ↳ System encrypts the attackers plaintext
- ↳ Attacker can pick patterns to encrypt that reveal the structure of the key.

3. Some concepts on Encryptions



4. Comparison between Link-to-Link and End-to-End Encryption

link to link	VS	End-to-End
→ message exposed in sending host & inter mediate nodes		→ message encrypted in sending host and receiving nodes.
→ requires one key per link pair		→ requires one key per processing pair
→ provides host/node authentication		→ provides application/user authentication
→ can be done in Hardware		→ usually done in software
→ applied by host as well as intermediate nodes		→ applied by sending host
→ More cipher text		→ Traffic analysis is easy.

5. Examples on calculating decryption time(s) for an exhaustive search using different key size (1/2)

Q: What an average time will be required for exhaustive search of a 32-bit, 56-bit and 128-bit AES algorithm on a machine that decrypts at 10^6 decryption/µsec.

Solution

32-bit AES Algorithm

→ Total no. of Keys $= 2^{32} = 4.3 \times 10^9$

→ Time required at 1 decryption/µsec $= \left(\dfrac{2^{31} \times 10^{-6}}{60 \times 60 \times 24 \times 365}\right)$ years

$= 6.81 \times 10^{-5}$ years

→ Time required at 10^6 decryption/µsec $= \left(\dfrac{6.81 \times 10^{-5}}{10^6}\right)$ years

$= 6.81 \times 10^{-11}$ years

→ (OPTIONAL) for conversion in minutes, multiply years with 525600

$= 6.81 \times 10^{-11} \times 525600 = 3.57 \times 10^{-5}$ min

$= 0.0000357$ minutes

5. Examples on calculating decryption time(s) for an exhaustive search using different key size (2/2)

H.W

Do same exhaustive search for 56-bit & 128-bit AES algo.

6. Symmetric Key Cryptography (1/8)

* Symmetric key encryption requires same key with the sender & receiver.
* Question: How to exchange key between two principals secretly?

Solution:

Diffie-Hellman (DH) Key exchange Algorithm

* DH uses a private-public key pair to establish a shared secret, typically a symmetric key. DH is not a symmetric algorithm — it is an asymmetric algorithm used to establish a shared secret for a symmetric key algorithm.
* Same key used for encryption/decryption
* Algorithm only for Key exchange & NOT for encryption/decryption.
* Based on discrete logarithm

6. Symmetric Key Cryptography (2/8)

Discrete Logarithms

→ If $\log_m b = a$, then $m^a = b$; where m = base. for e.g., $\log_{10}(100) = 2$, because $10^2 = 100$.

→ A discrete logarithm can be defined for integers only.

→ In fact we can define discrete logarithms mod p also where 'p' is any prime number.

Generator of Group

Examples

i) 2 is a generator of Z^*_{11} means: → Total 11 integers

$2^m \bmod 11$; where $m = 0, 1, 2, 3, 4, 5, \ldots 10$

↳ optional, can be ignored

optional field

m	0	1	2	3	4	5	6	7	8	9	10
$2^m \bmod 11$	1	2	4	8	5	10	9	7	3	6	1

$2^6 \bmod 11 = 64 \bmod 11 = 9$

6. Symmetric Key Cryptography (3/8)

Primitive Roots

* If $x^n = a$ then "a" is called the n^{th} root of x.

$$\sqrt[5]{243} = (243)^{1/5} = 3$$

for example:

if $3^5 = 243$, then $\sqrt[5]{243} = 3$

therefore we can say that 243 is the 5th root of 3 i.e $\overset{1}{3} \times \overset{2}{3} \times \overset{3}{3} \times \overset{4}{3} \times \overset{5}{3} = 243$

* $b = a^i \bmod p$

where b = any integer

a = primitive root

i = discrete logarithm

p = prime number

If $x^n = a$, then $\sqrt[n]{a} = x$

6. Symmetric Key Cryptography (4/8)

Example 1:

Is 3 a primitive root of prime number '7'?

Solution -

Given: $P = 7$

powers of 3 are:

$3^0 = 1 \bmod 7 = 1$

$3^1 = 3 \bmod 7 = 3$

$3^2 = 9 \bmod 7 = 2$

$3^3 = 27 \bmod 7 = 6$

$3^4 = 81 \bmod 7 = 4$

$3^5 = 243 \bmod 7 = 5$

$3^6 = 729 \bmod 7 = 1$

Repetition starts after 3^5 & it contains all integers before repetition, i.e., the integers/numbers b/w 1 & p-1; i.e, b/w 1 & 7-1 = 1 & 6 i.e, 1, 3, 2, 6, 4, 5 are b/w 1 & 6, therefore 3 is a primitive root of 7.

6. Symmetric Key Cryptography (5/8)

Example 2:

Is 2 a primitive root of prime number '7'?

Solution

$p = 7$

powers of 2 are:

$2^0 = 1 \bmod 7 = 1$

$2^1 = 2 \bmod 7 = 2$

$2^2 = 4 \bmod 7 = 4$

$2^3 = 8 \bmod 7 = 1$

$2^4 = 16 \bmod 7 = 2$

$2^5 = 32 \bmod 7 = 4$

$2^6 = 64 \bmod 7 = 1$

Although repetition starts after 2^2, but it doesn't contain all integers b/w 1 & p−1, i.e., b/w 1 & 6, 1, 2 & 4 are b/w 1 & 6 but 3, 5 & 6 are missing. Therefore 2 is not a primitive root of 7.

6. Symmetric Key Cryptography (6/8)

Diffie-Hellman Algorithm

Sparts:

1. Global public Elements
2. User A Public Key Generation
3. User B Public Key Generation
4. Generation of Secret key by user A.
5. Generation of Secret key by user B.

1. Global Public Elements:

↳ they are also called as domain parameters.

↳ p = prime number

↳ α; such that α < p & α is a primitive root of p.

⊗ for example: '3' is a primitive root of 7; here α = 3 & p = 7

6. Symmetric Key Cryptography (7/8)

2. **User A Public key Generation:**
   - ↳ Select private key "X_A" such that $X_A < P$
   - ↳ Calculate public key "Y_A" as: $\boxed{Y_A = \alpha^{X_A} \bmod p}$
3. **User B Public key Generation:**
   - ↳ select private key "X_B" such that $X_B < P$
   - ↳ Calculate public key "Y_B" as: $\boxed{Y_B = \alpha^{X_B} \bmod p}$

6. Symmetric Key Cryptography (8/8)

4. Generation of Secret key by user A:

$$\boxed{K_A = (Y_B)^{X_A} \bmod p}$$

5. Generation of secret key by user B:

$$\boxed{K_B = (Y_A)^{X_B} \bmod p}$$

$$\boxed{K_A = K_B = \text{symmetric key}}$$

7. Examples on Diffie-Hellman (1/13)

Example 1:

Given:

$P = 11$

$\alpha = 2$.

$\alpha < P$; $2 < 11$ (satisfied)

let's assume/choose $X_A = 4$ [$X_A < P$; $4 < 11$]

&

$X_B = 6$ [$X_B < P$; $6 < 11$]

User A public key generation

$Y_A = \alpha^{X_A} \bmod p = 2^4 \bmod 11 = 5$

$\boxed{Y_A = 5}$

7. Examples on Diffie-Hellman (2/13)

User B public key generation

$$Y_B = \alpha^{X_B} \bmod p = 2^6 \bmod 11 = 9$$

$$\boxed{Y_B = 9}$$

Generation of secret key by user A

$$K_A = (Y_B)^{X_A} \bmod p$$

$$= (9)^4 \bmod 11$$

$$\boxed{K_A = 5}$$

7. Examples on Diffie-Hellman (3/13)

Generation of secret key by user B

$$K_B = (Y_A)^{X_B} \bmod p$$

$$= (5)^6 \bmod 11$$

$$\boxed{K_B = 5}$$

$$\boxed{K_A = K_B = 5}$$

Home Work

Q: Perform/Apply Diffie-Hellman key exchange algorithm with $p = 23$ & $\alpha = 7$. Choose carefully the values of X_A & X_B and then compute values of Y_A & Y_B.

7. Examples on Diffie-Hellman (4/13)

Example 2:

Given:

$\alpha = 5$

$p = 97$

$\alpha < p$; $5 < 97$

→ We choose $X_A = 36$ & $X_B = 58$

$Y_A = \alpha^{X_A} \bmod p = 5^{36} \bmod 97$

$Y_A = 5^{36} \bmod 97$

* Many cryptographic procedures require modular exponentiation of large numbers to be carried out. A more appealing procedure known as Fast exponentiation will be used in this example.

→ $Y_A = 5^{36} \bmod 97$

7. Examples on Diffie-Hellman (5/13)

Where base = 5, exponent = 36

→ Convert exponent into its binary equivalent i.e

$(36)_{10} = 0\ 0\ 1\ 0\ 0\ 1\ 0\ 0_2$

→ The binary number contains 8 bits, therefore we will draw a table having 8 columns.

0	0	1	0	0	1	0	0
5	25	21	53	93	80	95	4

Answer

→ Write down each binary digit on the top of this column

7. Examples on Diffie-Hellman (6/13)

- **Write down the base which is '5' in 1st column of given table**

→ Now look at the binary digit of 2nd column which is zero in the given example.
If the binary digit is zero, we have to do only

i. If the binary digit is zero, we have to do only one step.

ii. If the binary digit is 1, you have to do two steps.

→ Take the number '5' from 1st column, square this number and modular 97 i.e.,

$5^2 \bmod 97 = 25$

Write this number '25' in 2nd column

→ Look at the binary digit at 3rd column in the table, which is one, therefore we have to do two steps:

i. $25^2 \bmod 97 = 43$

ii. $43 \times 5 \text{(base)} \bmod 97 = 21$

7. Examples on Diffie-Hellman (7/13)

→ Write this number '21' in 3rd column.

→ $21^2 \bmod 97 = 53$

→ $53^2 \bmod 97 = 93$

→ ↳ i → $93^2 \bmod 97 = 16$

↳ ii $16 \times 5 \bmod 97 = 80$

→ $80^2 \bmod 97 = 95$

→ $95^2 \bmod 97 = 4$

$Y_A = 5^{36} \bmod 97 = 4.$

$\boxed{Y_A = 4}$

7. Examples on Diffie-Hellman (8/13)

Now for $Y_B = \alpha^{X_B} \bmod p = 5^{58} \bmod 97$

$(58)_{10} = (00\ 11\ 10\ 10)_2$

c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8
0	0	1	1	1	0	1	0
5	25	21	71	82	31	52	85

$c_1 = 5$

$c_2 = 5^2 \bmod 97 = 25$

$c_3 =$
- $25^2 \bmod 97 = 43$
- $43 \times 5 \bmod 97 = 21$

$c_4 =$
- $21^2 \bmod 97 = 53$
- $53 \times 5 \bmod 97 = 71$

7. Examples on Diffie-Hellman (9/13)

$c_5 =$

- $71^2 \bmod 97 = 94$
- $94 \times 5 \bmod 97 = 82$

$c_6 = 82^2 \bmod 97 = 31$

$c_7 =$

- $31^2 \bmod 97 = 88$
- $88 \times 5 \bmod 97 = 52$

$c_8 = 52^2 \bmod 97 = 85$

$Y_B = 5^{58} \bmod 97 = c_8 = 85$; $\boxed{Y_B = 85}$

7. Examples on Diffie-Hellman (10/13)

Now for $K_A = (Y_B)^{X_A} \bmod p = (85)^{36} \bmod 97$

$$K_A = (85)^{36} \bmod 97$$

$$(36)_{10} = (0\ 0\ 1\ 0\quad 0\ 1\ 0\ 0)_2$$

c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8
0	0	1	0	0	1	0	0
85	47	70	50	75	12	47	75

$c_1 = 85$

$c_2 = 85^2 \bmod 97 = 47$

$c_3 =$

- $47^2 \bmod 97 = 75$
- $75 \times 85 \bmod 97 = 70$

$c_4 = 70^2 \bmod 97 = 50$

$c_5 = 50^2 \bmod 97 = 75$

7. Examples on Diffie-Hellman (11/13)

$C_6 =$
- $75^2 \bmod 97 = 96$
- $96 \times 85 \bmod 97 = 12$

$C_7 = 12^2 \bmod 97 = 47$

$C_8 = 47^2 \bmod 97 = 75$

$K_A = (85)^{36} \bmod 97 = C_8$

$$\boxed{K_A = 75}$$

7. Examples on Diffie-Hellman (12/13)

Now for $K_B = (Y_A)^{X_B} \bmod p = (4)^{58} \bmod 97$

$K_B = (4)^{58} \bmod 97$

$(58)_{10} = (0 0 1 1 1 0 1 0)_2$

c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8
0	0	1	1	1	0	1	0
4	16	54	24	73	91	47	75

$c_1 = 4$

$c_2 = 4^2 \bmod 97 = 16$

$c_3 = \cdot 16^2 \bmod 97 = 62$

$\quad \cdot 62 \times 4 \bmod 97 = 54$

$c_4 = \cdot 54^2 \bmod 97 = 6$

$\quad \cdot 6 \times 4 \bmod 97 = 24$

$c_5 = \cdot 24^2 \bmod 97 = 91$

$\quad \cdot 91 \times 4 \bmod 97 = 73$

7. Examples on Diffie-Hellman (13/13)

$c_6 = 73^2 \bmod 97 = 91$

$c_7 = 91^2 \bmod 97 = 36$

$\quad 36 \times 4 \bmod 97 = 47$

$c_8 = 47^2 \bmod 97 = 75$

$K_B = (4)^{58} \bmod 97 = c_8$

$\boxed{K_B = 75}$

$\boxed{K_A = K_B = 75}$

8. Asymmetric Key Operation (1/2)

* Asymmetric Key operation

→ A & B : principals (communicating parties)
T : Trusted third party

→ 'A' & 'B' do not have to approach 'T' jointly

→ 'B' approaches 'T' to obtain a key 'K_1' & sends key K_1 to 'A'.

→ 'B' also obtains key 'K_2' from 'T'

→ **K_1 & K_2 are different.**

→ B possesses a key pair (K_1, K_2)

K_1 → Public : Sent to A

K_2 → Private : from 'T'

8. Asymmetric Key Operation (2/2)

K_1 is used by A for locking / Encryption

K_1 available to general public $\Rightarrow$ Public key

K_2 is used by B for unlocking / Decryption

K_2 is held privately by B $\Rightarrow$ Private key

→ Always use public key for encryption & private key for decryption.

→ Public keys are available openly.

A (K_{PU}, K_{PR}) : Key pair

$$\underset{B}{\boxed{P.T}} \longrightarrow \boxed{E_{K_{PU}}(\;)} \xrightarrow[C.T]{\text{channel}} \underset{A}{\boxed{D_{K_{PR}}(\;)}} \longrightarrow \boxed{P.T}$$

References: Textbooks

1. *“Cryptography and Network Security”, by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2nd Edition, McGraw Hill Education (India) Private Limited.*
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